

Improving Diffusion Models with Optimal Diagonal Covariance Matching

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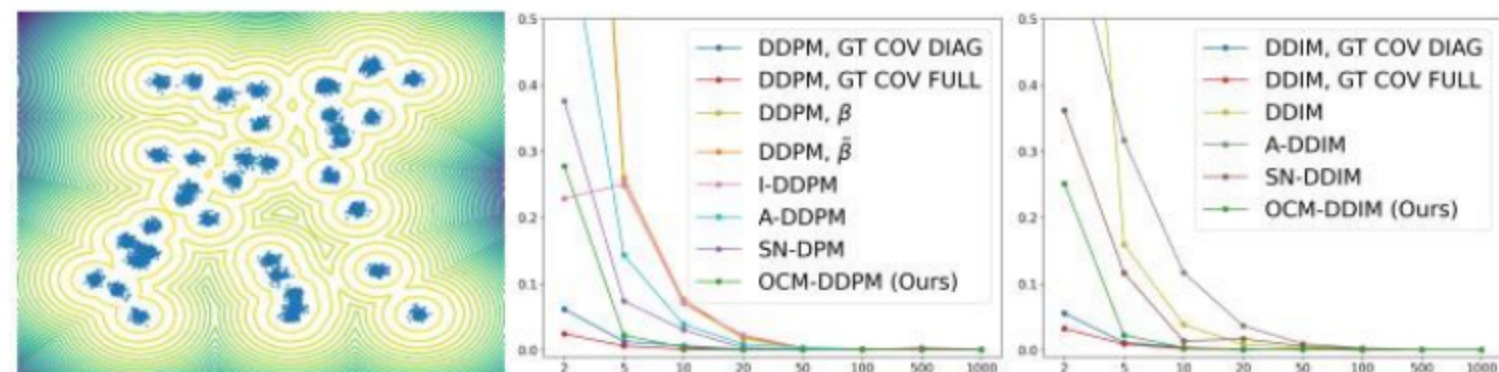
Sec 1

Introduction & Background

3 Motivation and Problem Statement

- Diffusion models face a quality-efficiency trade-off.
- Sampling covariance governs stochasticity and stability.
- Naïve diagonal estimates bias transitions and artifacts.
 - Ignoring off-diagonal structure omits key couplings.
 - Goal: optimally match diagonal covariances to data.
- Aim: boost likelihood and sample quality with speed.

2 DDPM vs DDIM: Essentials

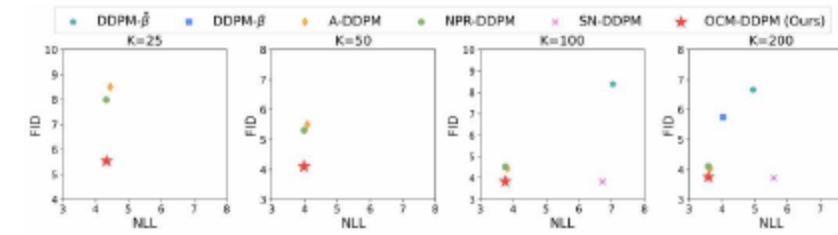


- DDPM: Markovian process with noisy, stochastic steps.
 - DDPM relies on heuristic choices for covariance.
 - DDIM: Non-Markovian with a deterministic trajectory.
- Accurate covariance boosts quality and enables faster sampling.

Sec 2

Optimal Covariance Matching (OCM)

6 Unbiased Diagonal Covariance Matching



- Unbiased objective for learning diagonal covariance
 - Targets accurate Hessian-diagonal estimation
 - Matches model covariance to a target distribution
- Diagonal choice balances fidelity and efficiency
 - Improves FID–NLL trade-off via covariance matching
 - Unbiased estimator cuts bias and improves accuracy

7 Training and Inference Details

- Objective combines score and diagonal covariance terms.
- Joint networks: score predictor plus Hessian estimator.
- K-step noise schedule trains across timesteps for stability.
 - Unbiased OCM estimator implemented with low overhead.
 - Inference uses predicted score and covariance to sample.
- Compatible with classifier-free guidance; stable outputs.

8 OCM Applications: DDPM vs DDIM



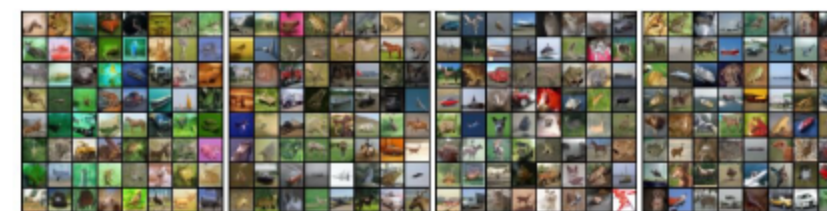
- OCM integrates with DDPM and DDIM seamlessly
 - Improves sample fidelity and recall across K
 - Fewer steps needed for similar visual quality
- Side-by-side grids compare methods over K
 - Guidance on choosing effective K per method
 - Enables acceleration without sacrificing quality

Sec 3

Experiments and Results

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Toy Demos & Hyperparameter Sensitivity



(a) OCM-DDPM ($M=5$) (b) OCM-DDPM ($M=10$) (c) OCM-DDPM ($M=15$) (d) OCM-DDPM ($M=20$)

- 2D Gaussian mixture toy task highlights OCM.
 - Varying M or K alters sample sharpness/diversity.
 - OCM outperforms baselines in covariance estimation.
- Accurate covariance drives cleaner synthesis.
 - Sensitivity analysis indicates strong stability.
 - Bias-variance behavior guides K and M choice.

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Image Modeling across Datasets



(e) CIFAR10 ($K=10$) (f) CelebA 64x64 ($K=17$) (g) LSUN Bedroom ($K=90$) (h) ImageNet 256x256 ($K=10$)

- OCM-DDPM matches or beats SOTA in FID and likelihood.
 - Improved FID-Recall balance with strong qualitative gains.
 - Consistent visual fidelity on CIFAR-10, CelebA, LSUN, ImageNet.
- Latent diffusion: faster, better quality, more diversity (DiT).

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Conclusion and Future Directions

- Unbiased OCM improves the likelihood-quality trade-off.
- Plug-and-play method compatible with DDPM and DDIM pipelines.
- Practical training and inference recipe for easy adoption.
 - Limitations and open questions outlined for future research.
 - Next steps: full covariance modeling and adaptive K /CFG strategies.
- Broaden to more modalities; acknowledge institutions and collaborators.